

# REFERENCE-FREE SINGING PITCH CORRECTION VIA MUSIC-CONSTRAINED SEQUENCE EDITING

Biao Dong<sup>1</sup>, Jiajun Li<sup>1</sup>, Binzhen Zhu<sup>1</sup>, Mingwei Yi<sup>1</sup>  
Tong Liu<sup>2</sup>, Yuanhao Zhang<sup>2</sup>, Jiqing Han<sup>1</sup>, Yongjun He<sup>1</sup>

<sup>1</sup>Faculty of Computing, Harbin Institute of Technology, Harbin, China

<sup>2</sup>China Unicom, Beijing, China

## ABSTRACT

Existing singing pitch correction approaches rely on target melodies or accompaniment tracks, which may be unavailable in practice. We formulate reference-free singing pitch correction as a music-constrained sequence editing task that determines whether and how each note should be corrected from the input performance alone. A pretrained symbolic music encoder with lightweight singing-domain adapters produces contextual representations of the singing MIDI. Based on these representations, two lightweight correction heads jointly model correction necessity and signed pitch modification through a factorized pitch-editing distribution. An input-dependent tonal prior derived from the estimated key distribution then reranks the candidate offsets, favoring tonally compatible corrections without target melodies or ground-truth key annotations. Experiments on real paired amateur and professional singing recordings show that the method improves note-level pitch accuracy from 73.25% to 83.43%, outperforming a context-based baseline by 3.99 percentage points while balancing error correction and preservation of correctly performed notes.

**Index Terms**— Singing pitch correction, reference-free, sequence editing, tonal constraint.

## 1. INTRODUCTION

Pitch accuracy is crucial for singing quality, yet amateur performances often exhibit significant pitch deviations that degrade overall musicality. Automatic singing pitch correction has become an important component in music production, online singing platforms, and AI-assisted music creation. However, many practical scenarios, including user-generated a cappella recordings, spontaneous humming, melody sketching, and legacy recordings without project files, provide only vocal performance without a reference melody. In such cases, reference-free correction must infer intended pitches from the musical context of the input while preserving expressiveness.

Existing singing pitch correction methods typically rely on explicit musical references or predefined pitch constraints. Commercial tools such as Auto-Tune and Melodyne provide scale- or note-constrained pitch editing [1, 2]. Academic systems likewise exploit auxiliary musical information. For instance, Deep Autotuner predicts note-wise pitch shifts from the vocal and accompaniment spectrograms [3], KaraTuner conditions pitch prediction on musical notes [4], and Neural Singing Voice Beautifier aligns an amateur performance with a professional template pitch curve [5]. More recently, generative models based on conditional variational autoencoders and diffusion have explored pitch correction under score- or MIDI-derived conditions [5, 6, 7, 8]. These systems assume a target

melody, accompaniment, professional template, or predefined scale. BERT-APC instead infers pitch corrections from the musical context of the input performance [9].

Reference-free singing pitch correction is fundamentally different from conventional pitch estimation. While pitch estimation focuses on extracting sung pitches, reference-free correction additionally requires determining whether each pitch should be modified. Similar edit decision formulations have been widely explored in sequence correction tasks, where error detection and editing operations are explicitly modeled for fine-grained sequence modification [10, 11, 12]. However, singing pitch correction introduces unique musical characteristics that differ from general sequence correction: the validity of a note depends not only on its pitch value but also on melodic context, tonal structure, and expressive singing characteristics. Music representation models underscore the importance of contextual modeling for music understanding [13], while singing research reveals interactions among pitch, technique, and expressive style [14]. We therefore model correction as note-level sequence editing informed by melodic context and tonality.

Based on these observations, we propose a reference-free singing pitch correction framework that formulates the task as a music-constrained sequence editing problem. Specifically, we adopt MusicBERT, a pretrained symbolic music representation model [15], as the backbone encoder and introduce lightweight singing-domain adapters to obtain contextual representations tailored to singing MIDI. Based on these representations, we design a hierarchical editing strategy with two lightweight correction heads that jointly model correction necessity and pitch modification, explicitly modeling note-level editing decisions. Furthermore, we estimate a tonal prior from the input singing performance and use it to rerank candidate corrections, introducing music-theoretic constraints without requiring target melodies or ground-truth key annotations. Audio examples are available at <https://dongb001.github.io/RF-SPC/>.

The main contributions of this work are as follows:

- We propose a reference-free singing pitch correction framework that models the musical context of the input performance using a frozen pretrained symbolic music encoder with lightweight singing-domain adapters.
- We develop a hierarchical correction model with two correction heads that jointly model correction necessity and pitch modification.
- We introduce tonal-aware candidate reranking using a tonal prior estimated from the input without ground-truth key annotations.

## 2. METHODOLOGY

### 2.1. Problem formulation

We formulate reference-free singing pitch correction in symbolic representation space. The correction pipeline is illustrated in Fig. 1. First, SOME [16] extracts note-level pitch, onset, and duration from an unaccompanied singing vocal recording, while RMVPE [17] extracts frame-level  $F_0$  for reconstruction. To construct the complete MusicBERT octuple representation, extracted pitch values are quantized to discrete semitones, the instrument is set to a fixed vocal code, and bar positions, time signature, and tempo are estimated from the onset sequence. After symbolic correction, modified pitches are mapped back to  $F_0$  contours via semitone-based pitch shifting on stable regions with linear interpolation at note boundaries, then resynthesized using NSF-HiFi-GAN [18, 19].

The extracted singing MIDI sequence is represented as

$$\mathcal{X} = \{x_i\}_{i=1}^N, \quad x_i = (p_i, d_i, s_i, e_i), \quad (1)$$

where  $N$  denotes the number of detected notes, and  $p_i, d_i, s_i$ , and  $e_i$  denote the pitch, duration, onset position, and ending position of the  $i$ -th note, respectively. Only pitch is edited while the temporal structure is preserved; for MusicBERT,  $\mathcal{X}$  is augmented with the remaining octuple fields to form  $\mathcal{X}_{\text{oct}}$ .

The objective is to generate a corrected singing representation

$$\mathcal{Y} = \{y_i\}_{i=1}^N, \quad y_i = (\hat{p}_i, d_i, s_i, e_i), \quad (2)$$

where  $y_i$  is the corrected note and  $\hat{p}_i$  is its pitch. The correction process can therefore be formulated as estimating a pitch modification  $\Delta_i = \hat{p}_i - p_i$  applied to the  $i$ -th note.

Since pitch errors are typically sparse in singing, an editing-based approach allows selective modification of erroneous notes while preserving correct ones, avoiding unnecessary alterations that full sequence prediction would introduce. The correction process consequently requires an editing decision that indicates whether a note should be modified. We introduce a binary editing indicator:

$$z_i = \begin{cases} 1, & \text{if note } i \text{ requires correction,} \\ 0, & \text{otherwise} \end{cases}. \quad (3)$$

The final corrected pitch is:

$$\hat{p}_i = p_i + \hat{\Delta}_i, \quad (4)$$

where  $\hat{\Delta}_i$  denotes the predicted pitch modification. A zero-valued prediction preserves the input pitch, whereas a nonzero prediction applies a correction. This formulation frames reference-free singing pitch correction as a music-constrained sequence editing task, where the model learns when and how to modify pitch deviations from the intrinsic musical context, guided by hierarchical editing decisions and tonal priors described in subsequent sections.

### 2.2. Singing-aware music context encoder

Reference-free singing pitch correction requires modeling musical context to infer appropriate pitch modifications. We therefore employ a pretrained symbolic music encoder to capture contextual dependencies among musical events. Specifically, we adopt MusicBERT as the backbone encoder [15], which learns contextual representations of structured musical events through Transformer-based pretraining on large-scale symbolic MIDI corpora.

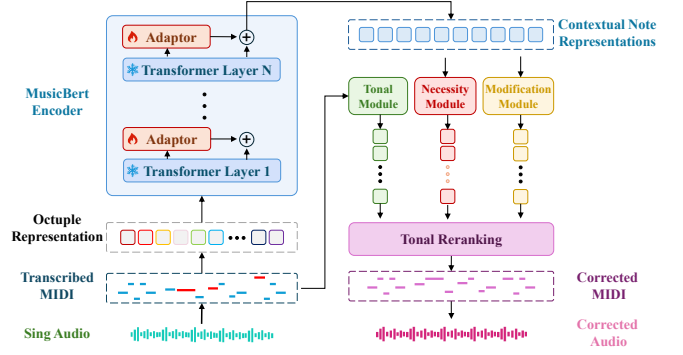


Fig. 1. Overview of the proposed reference-free singing pitch correction framework.

Although MusicBERT provides strong symbolic music representations, it is pretrained on general MIDI data, whereas singing MIDI exhibits different characteristics, including monophonic melodic structures and vocal-specific pitch variations. To bridge this domain gap, we introduce lightweight singing-domain adapters [20] while keeping the pretrained MusicBERT parameters frozen. The adapted representation is:

$$\tilde{H}_l = H_l + A_l(H_l), \quad l = 1, \dots, 12, \quad (5)$$

Here,  $H_l \in \mathbb{R}^{N \times 768}$  denotes the output of the  $l$ -th MusicBERT Transformer block, and  $h \in \mathbb{R}^{768}$  denotes an individual token representation. We use  $\tilde{H} = \tilde{H}_{12}$  as the final contextual representation. A gated adapter is inserted after each of MusicBERT's 12 Transformer blocks:

$$A_l(h) = \sigma(g_l) W_{2,l} \text{Drop}_{0.1}(\text{GELU}(W_{1,l} \text{LN}(h))), \quad (6)$$

where  $W_{1,l} \in \mathbb{R}^{64 \times 768}$ ,  $W_{2,l} \in \mathbb{R}^{768 \times 64}$ , and  $g_l$  is a trainable scalar gate for the adapter after the  $l$ -th Transformer block. The adapted representations are then fed into the subsequent hierarchical edit prediction module.

### 2.3. Hierarchical correction modeling

Inspired by explicit editing formulations in sequence correction tasks [11, 10], we formulate reference-free singing pitch correction as a hierarchical correction process. Instead of directly predicting the target pitch sequence, the proposed method decomposes correction into two stages: predicting correction necessity and predicting pitch modification. Given the contextual representation  $\tilde{H} = \{\tilde{h}_i\}_{i=1}^N$ , we introduce two lightweight correction heads, each consisting of a feed-forward projection layer followed by an activation function. The correction-necessity head estimates the probability that a note requires correction:

$$a_i = P(z_i = 1 | \tilde{h}_i) = \sigma(f_{\text{nec}}(\tilde{h}_i)). \quad (7)$$

The pitch-modification head predicts a distribution over the nonzero offset set  $\mathcal{D}_{\neq 0} = \{-3, -2, -1, +1, +2, +3\}$ , conditioned on correction:

$$b_i(d) = \text{Softmax}(f_{\text{mod}}(\tilde{h}_i))_d, \quad d \in \mathcal{D}_{\neq 0}. \quad (8)$$

The two predictions form a factorized pitch-editing distribution:

$$P(\Delta_i = d | \tilde{h}_i) = \begin{cases} 1 - a_i, & d = 0, \\ a_i b_i(d), & d \neq 0. \end{cases} \quad (9)$$

The two correction heads are jointly optimized with classification losses:

$$\mathcal{L} = \mathcal{L}_{\text{nec}} + \mathcal{L}_{\text{mod}}, \quad (10)$$

where  $\mathcal{L}_{\text{nec}}$  and  $\mathcal{L}_{\text{mod}}$  are defined as

$$\mathcal{L}_{\text{nec}} = -\frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} [z_i^* \log a_i + (1 - z_i^*) \log(1 - a_i)], \quad (11)$$

$$\mathcal{L}_{\text{mod}} = -\frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \log b_i(\delta_i^*). \quad (12)$$

Here  $\mathcal{V}$  denotes the set of valid notes,  $\mathcal{E} \subseteq \mathcal{V}$  denotes the subset of notes that were perturbed in the training data,  $z_i^* = \mathbf{1}[i \in \mathcal{E}]$  is the binary correction label, and  $\delta_i^*$  is the target offset for note  $i \in \mathcal{E}$ . The necessity prediction is trained with binary cross-entropy over all notes, whereas the modification prediction is optimized with cross-entropy only on notes that require correction.

#### 2.4. Tonal-aware reranking

Although the hierarchical correction model predicts pitch modifications from singing context, reference-free correction can still yield multiple plausible candidates without an explicit melody reference. We therefore introduce a tonal-aware candidate reranking strategy. Rather than imposing a hard key constraint, it uses tonal information as a soft prior derived solely from the input singing performance.

We apply the Krumhansl–Schmuckler (KS) key-finding algorithm [21] to the input sequence  $\mathcal{X}$ . Let  $u_{\mathcal{X}}$  be its duration-weighted 12-bin pitch-class histogram, whose  $v$ -th bin is  $u_{\mathcal{X}}(v) = \sum_i d_i \mathbf{1}[(p_i \bmod 12) = v]$ . Pearson correlations with the 24 major and minor key templates  $\kappa_k$  form a soft key posterior:

$$\rho(k | \mathcal{X}) = \text{Softmax}_k(\text{corr}(u_{\mathcal{X}}, \kappa_k)/\tau). \quad (13)$$

We denote the resulting posterior vector by

$$\rho_{\mathcal{X}} = \{\rho(k | \mathcal{X})\}_{k=1}^{24}.$$

We set the posterior temperature to  $\tau = 0.15$ . The posterior induces an input-dependent pitch-class prior:

$$r_{\mathcal{X}}(v) = \sum_k \rho(k | \mathcal{X}) \pi_k(v), \quad (14)$$

where  $v \in \{0, \dots, 11\}$ . For the seven-class scale set  $\mathcal{S}_k$ ,  $\pi_k(v)$  assigns weight 1 to  $v \in \mathcal{S}_k$  and  $\eta = 0.15$  otherwise, normalized by  $7 + 5\eta$ . We compute a song-level confidence  $c(\mathcal{X}) \in [0, 1]$  from the posterior entropy and the input-note count  $N$ :

$$c(\mathcal{X}) = \left(1 - \frac{\mathcal{H}(\rho_{\mathcal{X}})}{\log 24}\right) \min\left(1, \frac{N}{2n_{\min}}\right), \quad (15)$$

where  $\mathcal{H}(\rho_{\mathcal{X}}) = -\sum_k \rho(k | \mathcal{X}) \log \rho(k | \mathcal{X})$ . We set  $n_{\min} = 8$ ; when  $N < n_{\min}$  or  $u_{\mathcal{X}}$  has zero variance, we use a uniform posterior and set  $c(\mathcal{X}) = 0$ . For each note, the editor considers the seven offsets  $\mathcal{D} = \{0, \pm 1, \pm 2, \pm 3\}$ , where 0 indicates no modification. We combine the editing distribution with the tonal prior using:

$$S_i(d) = \log P(\Delta_i = d | \tilde{h}_i) + \lambda c(\mathcal{X}) \log r_{\mathcal{X}}((p_i + d) \bmod 12), \quad (16)$$

where  $\lambda \geq 0$  controls the global strength of the tonal constraint and is selected on the validation set. The final pitch modification is:

$$\hat{\Delta}_i = \arg \max_{d \in \mathcal{D}} S_i(d). \quad (17)$$

The corresponding predicted editing decision is  $\hat{z}_i = \mathbf{1}[\hat{\Delta}_i \neq 0]$ . This confidence-weighted reranking biases ambiguous candidates toward tonally compatible solutions without forcing pitches to conform to a single detected key.

### 3. EXPERIMENTAL SETUP

#### 3.1. Dataset

We construct training samples by extracting singing MIDI from 16,000 professional recordings (approximately 1,060 hours of Chinese and English popular songs) and applying synthetic pitch perturbations. Considering the typical vocal range, note pitches are constrained within MIDI 36–84 (C2–C6). For each sequence, 20% of notes are perturbed using the six nonzero offsets in  $\mathcal{D}_{\neq 0}$ . To simulate different error patterns, 70% of perturbations are independent, whereas 30% form correlated spans of two to four notes. Synthetic perturbations are used for model development, and real paired data are reserved exclusively for evaluation.

For evaluation, we use a separate private paired singing dataset containing Chinese and English popular songs. Each song includes amateur and professional performances recorded by vocal students. The paired recordings are aligned using dynamic time warping (DTW) to obtain note-level correspondence. The dataset is split by song, including 20 validation songs (8,299 notes) and 80 held-out test songs (30,682 notes). All hyperparameters, including the tonal reranking strength, are selected using only the validation set. The professional performances are used as evaluation references rather than absolute ground truth, since expressive variations may exist across performances.

#### 3.2. Implementation Details

The pretrained MusicBERT encoder remains frozen throughout training. We use three training stages: (1) train only the adapters for 8k updates with MusicBERT’s original masked-token prediction head; (2) train only the two correction heads for 12k updates; and (3) jointly train the adapters and correction heads for 25k updates. The learning rate is  $10^{-4}$  in Stages 1 and 2; Stage 3 uses  $10^{-5}$  for adapters and  $10^{-4}$  for correction heads. We use the AdamW optimizer with cosine annealing schedules. The correction necessity and pitch modification losses are equally weighted. Each GPU processes 32 sequences with a maximum length of 8,192 field tokens.

#### 3.3. Baselines

We compare our method with two reference-free baselines: an independent reimplementation of BERT-APC [9] trained on the same corpus, and a key-based pitch snapping method that adjusts notes toward the nearest pitch classes according to the estimated key. BERT-APC is a recent learning-based approach to this task.

#### 3.4. Evaluation Metrics

We report note-level semitone-accurate pitch accuracy (Acc.) as the primary metric. To evaluate correction-preservation trade-offs, we report the error repair rate ( $N_{\text{fixed}}/N_{\text{originally wrong}}$ ) and harm rate ( $N_{\text{damaged}}/N_{\text{originally correct}}$ ), measuring recovery of incorrect notes and undesired modification of correct notes, respectively. For subjective evaluation, we assess three perceptual aspects on a five-point Likert scale: Pitch Correctness (pitch accuracy relative to piano-rendered reference recordings), Naturalness (vocal quality preservation), and Overall Quality (listening preference).

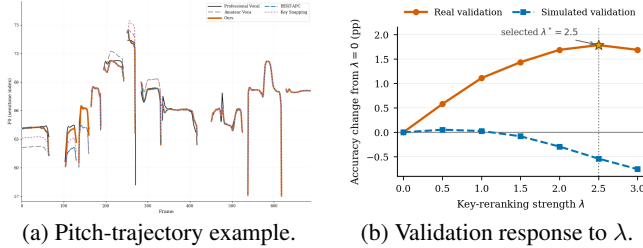


Fig. 2. Qualitative and validation analyses.

## 4. RESULTS

We first evaluate overall correction performance, then examine component contributions, and finally assess perceptual quality.

### 4.1. Objective evaluation

Table 1 compares the proposed method with two reference-free baselines on the held-out real test set. The proposed method improves note-level pitch accuracy from 73.25% to 83.43%, increasing agreement with the professional references on the held-out paired recordings. It exceeds BERT-APC by 3.99 percentage points, and both gains over the correction baselines are significant (paired song-level Wilcoxon tests; Holm-adjusted  $p < 0.001$ ). The proposed method also raises the error repair rate from 26.18% to 46.81%, while its harm rate remains 3.19%. Key Snapping repairs 20.76% of the originally incorrect notes but damages 9.62% of the correct notes, reducing accuracy below the uncorrected input. A representative pitch-trajectory example is shown in Fig. 2(a).

Table 1. Objective results on the held-out real test set (%).

| Method       | Acc. $\uparrow$ | $\Delta$ Acc. $\uparrow$ | Rep. $\uparrow$ | Harm $\downarrow$ |
|--------------|-----------------|--------------------------|-----------------|-------------------|
| Input        | 73.25           | 0.00                     | —               | —                 |
| Key Snapping | 71.76           | -1.49                    | 20.76           | 9.62              |
| BERT-APC [9] | 79.44           | +6.19                    | 26.18           | <b>1.11</b>       |
| <b>Ours</b>  | <b>83.43</b>    | <b>+10.18</b>            | <b>46.81</b>    | 3.19              |

### 4.2. Ablation study

To isolate the contribution of each component, we evaluate variants without singing-domain adaptation, hierarchical correction modeling, or tonal-aware reranking. Table 2 reports the results on the held-out real test set.

**Singing-domain adaptation:** We remove the adapter modules and optimize only the correction heads on top of frozen MusicBERT representations. Removing the adapters reduces note accuracy from 83.43% to 76.87%, showing that adaptation to singing-specific MIDI characteristics is important for reference-free pitch correction.

**Hierarchical correction modeling:** The flat edit head variant replaces the two correction heads with a single seven-way classifier. Compared with the flat head, the hierarchical formulation increases the repair rate from 39.32% to 46.81% and accuracy from 82.35% to 83.43%, at the cost of a higher harm rate (3.19% vs. 1.93%).

**Tonal-aware reranking:** To evaluate the contribution of tonal information, we remove the tonal-aware reranking module and compare with the proposed method. Removing tonal reranking decreases note-level accuracy from 83.43% to 81.52% and the error repair

Table 2. Ablation results on the held-out real test set (%).

| Method         | Acc. $\uparrow$ | $\Delta$ Acc. $\uparrow$ | Rep. $\uparrow$ | Harm $\downarrow$ |
|----------------|-----------------|--------------------------|-----------------|-------------------|
| w/o adapters   | 76.87           | +3.62                    | 34.10           | 7.52              |
| Flat edit head | 82.35           | +9.10                    | 39.32           | 1.93              |
| w/o reranking  | 81.52           | +8.27                    | 35.76           | <b>1.77</b>       |
| <b>Ours</b>    | <b>83.43</b>    | <b>+10.18</b>            | <b>46.81</b>    | 3.19              |

rate from 46.81% to 35.76%. This indicates that tonal information provides useful musical priors for selecting correction candidates. We further analyze the influence of tonal-constraint strength  $\lambda$  on real paired singing data, as shown in Fig. 2(b). Tonal constraints provide limited improvement on synthetic perturbations, where randomly generated pitch errors are less correlated with musical structure. In contrast, real singing performances benefit more from tonal information, indicating that realistic singing deviations have stronger relationships with the underlying tonal context. The highest real-validation accuracy is obtained at  $\lambda = 2.5$ . Accuracy decreases when the tonal term is weighted more strongly, suggesting that an overly large weight may overemphasize the estimated tonal prior. This observation motivates the use of a moderately weighted soft tonal prior.

### 4.3. Subjective evaluation

To evaluate the perceptual quality of the corrected singing audio, we conduct a subjective listening test with 30 vocally trained participants. Each participant evaluates 5 randomly selected singing excerpts (8–12 seconds) across four methods. Participants are provided with piano-rendered professional recordings as pitch references before rating. All methods use identical SOME transcription and NSF-HiFi-GAN reconstruction; only symbolic correction differs.

As shown in Table 3, the proposed method achieves the highest mean scores for Pitch Correctness and Overall Quality, improving them from 3.10 to 3.59 and from 3.25 to 3.94 over the input, respectively. Its Naturalness score remains close to the input (3.93 vs. 3.97), indicating that the perceptual improvements are achieved without substantially degrading naturalness.

Table 3. Subjective evaluation on a five-point scale.

| Method       | Pitch Corr. $\uparrow$ | Nat. $\uparrow$ | Overall $\uparrow$ |
|--------------|------------------------|-----------------|--------------------|
| Input        | 3.10                   | <b>3.97</b>     | 3.25               |
| Key Snapping | 3.26                   | 3.80            | 3.67               |
| BERT-APC     | 3.46                   | 3.85            | 3.47               |
| <b>Ours</b>  | <b>3.59</b>            | 3.93            | <b>3.94</b>        |

## 5. CONCLUSIONS

We formulate reference-free singing pitch correction as a music-constrained sequence editing task by combining a pretrained symbolic music encoder with lightweight singing-domain adapters, hierarchical correction modeling, and tonal-aware reranking. Evaluations on real paired amateur and professional recordings show improved pitch correction while largely preserving naturalness. Due to the inherent diversity of musical expression, reference-free correction produces one musically plausible interpretation among multiple valid alternatives. Future work will extend the framework to jointly model pitch, rhythm, and singing techniques for more comprehensive singing voice refinement.

## 6. REFERENCES

- [1] Harold A. Hildebrand, “Pitch detection and intonation correction apparatus and method,” U.S. Patent 5,973,252, October 1999.
- [2] Celemony Software GmbH, “What is Melodyne? note-based audio editing,” Online technical documentation, Accessed: Aug. 22, 2026.
- [3] Sanna Wager, George Tzanetakis, Cheng-i Wang, and Minje Kim, “Deep Autotuner: A pitch correcting network for singing performances,” in *ICASSP 2020–2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 246–250.
- [4] Xiaobin Zhuang, Huiran Yu, Weifeng Zhao, Tao Jiang, and Peng Hu, “KaraTuner: Towards end-to-end natural pitch correction for singing voice in karaoke,” 2022.
- [5] Jinglin Liu, Chengxi Li, Yi Ren, Zhiying Zhu, and Zhou Zhao, “Learning the beauty in songs: Neural singing voice beautifier,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 7970–7983, Association for Computational Linguistics.
- [6] Jiarui Hai and Mounya Elhilali, “Diff-Pitcher: Diffusion-based singing voice pitch correction,” in *2023 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2023, pp. 1–5.
- [7] Jianzong Wang, Pengcheng Li, Xulong Zhang, Ning Cheng, and Jing Xiao, “ConTuner: Singing voice beautifying with pitch and expressiveness condition,” in *2024 International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–6.
- [8] Wenhui Li, Biao Dong, Liwei Hu, Jiqing Han, and Yongjun He, “SRF-SVB: Style-consistent singing voice beautifying via rectified flow,” 2026.
- [9] Sungjae Kim, Kihyun Na, Jinyoung Choi, and Injung Kim, “BERT-APC: A reference-free framework for automatic pitch correction via musical context inference,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 2649–2662, 2026.
- [10] Kostiantyn Omelianchuk, Vitaliy Atrasevych, Artem Chernodub, and Oleksandr Skurzhashnyi, “GECToR—grammatical error correction: Tag, not rewrite,” in *Proceedings of the Fifteenth Workshop on Innovative Use of NLP for Building Educational Applications*, 2020, pp. 163–170, Association for Computational Linguistics.
- [11] Eric Malmi, Sebastian Krause, Sascha Rothe, Daniil Mirylenka, and Aliaksei Severyn, “Encode, tag, realize: High-precision text editing,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 5054–5065, Association for Computational Linguistics.
- [12] Jonathan Mallinson, Aliaksei Severyn, Eric Malmi, and Guillermo Garrido, “FELIX: Flexible text editing through tagging and insertion,” in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1244–1255, Association for Computational Linguistics.
- [13] Yizhi Li, Ruibin Yuan, Ge Zhang, Yinghao Ma, Xingran Chen, Hanzhi Yin, Chenghao Xiao, Chenghua Lin, Anton Ragni, Emmanouil Benetos, Norbert Gyenge, Roger Dannenberg, Ruibo Liu, Wenhui Chen, Gus Xia, Yemin Shi, Wenhao Huang, Zili Wang, Yike Guo, and Jie Fu, “MERT: Acoustic music understanding model with large-scale self-supervised training,” in *International Conference on Learning Representations*, 2024, pp. 12181–12204.
- [14] Yu Zhang, Changhao Pan, Wenxiang Guo, Ruiqi Li, Zhiyuan Zhu, Jialei Wang, Wenhao Xu, Jingyu Lu, Zhiqing Hong, Chuxin Wang, Lichao Zhang, Jinzheng He, Ziyue Jiang, Yuxin Chen, Chen Yang, Jiecheng Zhou, Xinyu Cheng, and Zhou Zhao, “GTSinger: A global multi-technique singing corpus with realistic music scores for all singing tasks,” in *Advances in Neural Information Processing Systems 37*, 2024, pp. 1117–1140.
- [15] Mingliang Zeng, Xu Tan, Rui Wang, Zeqian Ju, Tao Qin, and Tie-Yan Liu, “MusicBERT: Symbolic music understanding with large-scale pre-training,” in *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2021, pp. 791–800, Association for Computational Linguistics.
- [16] OpenVPI, “SOME: Singing-oriented MIDI extractor,” GitHub repository, Accessed: Sep. 24, 2026.
- [17] Haojie Wei, Xueke Cao, Tangpeng Dan, and Yueguo Chen, “RMVPE: A robust model for vocal pitch estimation in polyphonic music,” in *Interspeech 2023*, 2023, pp. 5421–5425.
- [18] Xin Wang, Shinji Takaki, and Junichi Yamagishi, “Neural source-filter waveform models for statistical parametric speech synthesis,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 402–415, 2020.
- [19] Jungil Kong, Jaehyeon Kim, and Jaekyoung Bae, “HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis,” in *Advances in Neural Information Processing Systems*, 2020, vol. 33, pp. 17022–17033.
- [20] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly, “Parameter-efficient transfer learning for NLP,” in *Proceedings of the 36th International Conference on Machine Learning*, 2019, vol. 97 of *Proceedings of Machine Learning Research*, pp. 2790–2799, PMLR.
- [21] Carol L. Krumhansl, *Cognitive Foundations of Musical Pitch*, Oxford University Press, Nov. 2001.